

Adversarial Methodology Review — Round 3 (2026-07-09)

Scope. One further adversarial round on (a) the underlying flood-model methodology and (b) the claims in docs/paper/ieee-kse2026.tex, with specific attention to *why each approach was chosen* and to *heterogeneity across the four cities* and its justification. Builds on Round 1

(scripts/_referee_stress_test.py: exact Fisher/Clopper-Pearson statistics, hazard-layer ablations, control-difficulty audit) and Round 2 (scripts/_referee_stress_test2.py: elevation-null baseline, Jakarta handfill-stage sensitivity). This round covers what those did not: the forcing chain, coastal-solver design choices, the cross-city configuration register, and paper-code consistency. Every factual claim below was verified against the code, the release script (repro/run_atlas_fixed.sh), or a measured output; nothing is argued from memory.

Verdict key. DEFENSIBLE — choice is anchored and documented; no action. DISCLOSED — defensible but only because the limitation is (now) stated; keep disclosure. FIXED — challenge found a wrong or stale claim; corrected this round. OPEN — defensible as screening-grade, but a named follow-up test would materially strengthen it.

1. Summary of outcomes

#	Challenge	Verdict
C 1	Mitigation-delta mechanism in the paper ("difference of two inflated extents is itself inflated")	FIXED — claim was empirically false
C 2	Abstract: DSM high bias "drives coastal over-prediction"	FIXED — wrong mechanism
C 3	Limitations: pluvial "heterogeneous... explains residual specificity loss"	FIXED — stale (pre-dated regime-matching)
C 4	Methods: KL described as handfill-only	FIXED — released KL layer is a union
C 5	Uniform coastal Manning $n = 0.06$, hardcoded	OPEN — sensitivity untested
C 6	Synthetic 3-1-2 h surge hydrograph + $t_{\text{end}} = 8$ h	OPEN — duration sensitivity untested
C 7	Canal-HAND stages/pruning chosen with gate visibility	DISCLOSED — single-iteration risk stated; held-out event test recommended
C 8	Bathtub run-flag --fluvial-bankfull-rp 0 vs paper's "bankfull subtracted"	DEFENSIBLE — subtraction is baked upstream (audit-trap documented below)
C 9	Pumped-polder floor applied to Bangkok only	DEFENSIBLE — rationale now written down (was code-comment only)
C 10	Bare-earth vs FABDEM	DEFENSIBLE — licence-driven, per-city lidar-fit
C 11	Singapore DeltaDTM hybrid (only city not lidar-calibrated)	DISCLOSED
C 12	GEV shape clamp $\xi \leq 0.30$; RP1000 extrapolation; single gauge per city	DISCLOSED in SAFE; KSE carries no forcing-uncertainty sentence (space) — noted
C 13	Bangkok "significant" on a 7-control degenerate stratum (Fisher ≈ 0.08)	DISCLOSED — criterion named inline; keep both statistics
C 14	IDF anchor duration 1 h (SG) vs 6 h (others)	DEFENSIBLE — service-standard-matched; KL 1-h variant untested
C 15	Scoring all registers at RP100 ("event-matched")	DEFENSIBLE — symmetric footprint penalises over-flooding; wording could be tighter

2. Detailed challenges, rationale, and dispositions

2.1 Paper-claim challenges (all four FIXED in `ieee-kse2026.tex` this round)

C1 — The mitigation-delta mechanism was empirically false. *Claim (before):* "the bathtub bias is multiplicative ($5.2\times$), so a difference of two inflated extents is itself inflated." *Challenge:* we measured it. Engine bathtub on the released seawall DEM at 2100: $SSP2-4.5 = 4,166 \text{ km}^2$, $SSP5-8.5 = 4,201 \text{ km}^2$ → avoided area $\approx 35 \text{ km}^2$. Inertial: 760.5 vs 852.8 km^2 → **92 km²**. The bathtub *deflates* the delta ($0.4\times$), it does not inflate it: the static fill saturates the flat delta under both high-SLR pathways and washes out the scenario signal. The conclusion ("read the delta off the inertial layer") survives; the stated mechanism reversed. *Why the wrong claim existed:* it was inherited from the DSM-era numbers (-133 vs $\approx 11 \text{ km}^2$), where the un-saturated building-blocked bathtub did inflate the delta. On bare earth the regime changed and the sentence was not re-derived. *Fix:* paragraph rewritten with the measured 35-vs-92 saturation mechanism. Provenance: `outputs/_bathtub_check/bkk2100_src_polder, bkk245_src_polder; repro/bathtub_bias_check.sh` (2100 block).

C2 — Abstract attributed coastal over-prediction to the DSM's high bias. *Challenge:* measured bathtub extents are *larger* on bare earth than on the DSM (Bangkok $4,432$ vs $3,546 \text{ km}^2$) — building mass acted as an accidental barrier, so the DSM bias *masked* flooding rather than driving over-prediction; the over-prediction is the solver's. The paper's own §DEM sentence ("what makes the bathtub over-prediction a solver-architecture artifact rather than a terrain artifact") already said this correctly; the abstract contradicted it. *Fix:* abstract clause now reads "...high bias that masks documented flood locations" (the gate-evidenced mechanism: the DSM masked Old Klang Road and canal corridors).

C3 — Limitations carried a stale pluvial sentence. "Currently heterogeneous (a documented homogenisation step)... explains residual specificity loss from over-ponding on elevated ground" described the pre-regime-matching state and the old Jakarta fill-spill CRR failure; current CRRs are 0.90 – 1.00 and the split is deliberate. *Fix:* reworded to regime-matched-by-mechanism with the actual residual cost (CRR 0.92 in JKT/KL where the landfill over-reaches).

C4 — Methods under-described the released KL pluvial. The released layer is `max(fill-spill, canal-HAND)` (`scripts/_apply_kl_canal_pluvial.py`), not landfill alone; Jakarta's is `max(engine landfill, canal-HAND)`. *Fix:* parenthetical added (KL unioned per-pixel with its depression cascade). Reproducibility chain for the unions: `repro/run_atlas_fixed.sh` → `repro/rerun_handfill_scaled.sh` → `scripts/_apply_{kl,jakarta}_canal_pluvial.py` (run docs: `docs/runs/2026-07-04-*.md`).

2.2 Coastal solver

Why local-inertia at all. Bates et al. (2010) reduced-physics SWE: drops advection (valid $Fr \ll 1$ — surge/backwater regime), semi-implicit friction, explicit continuity, CFL-adaptive Δt on a staggered grid, numba-JIT with a verified numpy fallback. Chosen over (a) bathtub — no dynamics, measured $5.2\times/1.2\times$ over-prediction on identical terrain; (b) full SWE (e.g. ANUGA/LISFLOOD-FP full solvers) — cost and licence-free-stack constraints; the local-inertia family is the standard screening compromise (LISFLOOD-FP's own default). The $5.2\times$ Bangkok / $1.2\times$ Jakarta comparison is apples-to-apples by construction (same seawall DEM, sea mask, tidal seeds, water level; only `--coastal-solver` differs) — re-verified this week via `repro/bathtub_bias_check.sh`.

C5 — Uniform Manning $n = 0.06$ (hardcoded at the `run_inertial` call). A mangrove fringe and a paved promenade get identical friction, and n directly controls inland propagation distance. The *pluvial* rain-on-grid solver takes a per-cell n raster; the coastal solver never does — an internal inconsistency of sophistication. *Defense:* 0.06 is a mid-range urban-floodplain value; screening grade; extent is reported at a 0.10 m threshold where first-order geometry dominates. *Disposition:* OPEN — run the $n \in \{0.03, 0.10\}$ bracket for one city (Bangkok 2100 RP100 is the sensitive case) and record the extent range in the parameter register. Until then this is undisclosed in the paper.

C6 — Idealised surge forcing. One synthetic triangular hydrograph (3 h ramp / 1 h hold / 2 h recession to a permanent MSL+SLR floor) for every city, RP, and scenario; $t_{\text{end}} = 8 \text{ h}$; convergence deliberately disabled when the SLR floor is active. *Why:* a generic design event (no per-storm reconstruction is possible for a 5-scenario $\times$ 3-RP atlas); the asymmetric ramp is a numerical-stability

requirement (Dirichlet shock on a dry bed); the permanent floor is what makes SLR inundation persist after the transient surge — recognised and handled. *Challenge*: inundation extent is then surge-duration-limited by construction; if real Gulf-of-Thailand surges persist longer than 6 h, the inertial extent is under-resolved, and part of the 5.2× bathtub gap would be "finite-duration physics" (defensible, indeed the point) mixed with "t_end truncation" (numerical). The subsea pre-flood + permanent floor mean the *SLR component* is not duration-limited; the *surge component* is. *Disposition*: OPEN — a single t_end-doubling run (16 h) for Bangkok 2100 would bound the truncation term; recommend recording converged vs t_end-limited per released coastal cell (the solver already returns it).

Datum chain (C12-adjacent). Gauge de-means to local MSL → CMEMS CNES-CLS-2022 MDT offset → EGM2008 (per-city: SG 1.1588 m, JKT 0.9976 m, BKK 1.1785 m). Best available open practice; dm-scale MDT error propagates 1:1 into level on flat deltas — inherent to any open datum chain, disclosed in the docs. The GEV shape clamp $\xi \in [-0.5, 0.30]$ prevents unstable Fréchet fits on 30–50-yr records; the clamp was validated against PUB ponding ranges for the *pluvial* fit and is asserted, not validated, for surge — RP1000 coastal is a deep extrapolation either way (SAFE v5 discloses this; KSE omits the forcing-uncertainty sentence for space — acceptable for a 6-page venue, noted here).

2.3 Fluvial

Why GloFAS → Manning rating → main-stem HAND. No licensed hydrodynamic model fits the open-stack constraint; GloFAS provides the only free design discharge. The 180 km² main-stem threshold is anchored to *which channel the modelled discharge physically represents* (fixed before the first gate run; corrected a false positive on a 60–77 m hill), and its failure boundary (out-of-domain mega-rivers: Chao Phraya, Ciliwung upstream) is disclosed as contribution #5 rather than hidden.

C8 — the bankfull audit-trap (verified consistent). The release script passes `--fluvial-bankfull-rp 0`, which *looks like* it disables the bankfull subtraction the paper claims. It does not: the subtraction is baked upstream in `scripts/fit_fluvial_glofas.py` (Manning stage at `Q_bankfull` subtracted from every RP stage before the hazard CSV is written — e.g. Bangkok RP100 fluvial `water_level_m = 0.61` m, an overbank *residual*, not a full stage). The run-time flag is 0 precisely to avoid double-subtraction. **Any auditor diffing the release script against the paper will hit this**; it is now documented here and in §10 of the model documentation.

2.4 Pluvial

Why regime-matching instead of one scheme. Fill-and-spill answers "where does rain pond?" (depression storage — Bangkok's flat micro-topography); canal-HAND handfill answers "where does the drainage network overflow?" (canal-dense KL/SG/JKT, where the documented flood record is drainage-corridor flow that 30 m depression storage cannot see — the hazard-ablation in Round 1 measured exactly this). A single scheme would either speckle the canal cities (fill-spill) or invent corridors in Bangkok (handfill). The cost — heterogeneity — is now framed as a deliberate, mechanism-anchored split (C3 fix).

C7 — how much of the canal-HAND configuration saw the gate? The honest accounting: the KL pruning rule (drains ≥ 0.5 km² accumulation) is physically anchored (headwater ditches on hills are not flood conveyance) and Jakarta was shown gate-*insensitive* to pruning (all four variants scored identically — `_apply_jakarta_canal_pluvial.py`); but the stage scalings (KL 0.5 m at 0.095 m reference excess, clip [0.4, 2.5]; JKT 0.3 m at 0.130 m, clip [0.15, 1.5]; SG engine stage 1.5 m, cap 0.5 m) were selected with gate visibility, and KL's PASS is the direct product of one model-improvement iteration against a frozen register (HR 0.65 → 0.71; TSS 0.65 → 0.63, i.e. parity within the ± 0.16 CI). *Defense in place*: register frozen before any scoring and never re-labelled; the § footnote in the paper discloses the conditionality and the revert delta; Round-2's elevation-null showed the KL register itself cannot fully distinguish model from topography (null TSS 0.92), which the SAFE paper discloses as "necessary, not sufficient." *Disposition*: DISCLOSED, with the standing recommendation unchanged from `model-documentation.md` §10 item 3: physically re-anchor the base stages (drain design freeboard), and validate the canal-HAND layer against a **held-out event** (a post-2022 flood not in any register) before the next submission cycle.

C14 — IDF anchor duration heterogeneity (1 h SG vs 6 h others). Matched to each service's published design practice (PUB CoP sub-hourly convective design for the secondary drainage system; MSMA/TMD-RID/BMKG 6-h standards). *Residual challenge:* KL's convective bursts are also sub-hourly; a 1-h KL variant was never run. Defensible as standard-anchored (the December-2021 KL event was long-duration), but the sensitivity is untested — logged.

Depth caps (0.5 / 1.0 / 3.0 / 3.0 m for SG/JKT/KL/BKK). Anchored respectively to the PUB observed ponding range (0.07–0.76 m), documented Jakarta kampung inundation depths, and (KL/BKK) an effectively-uncapped life-safety bound where the fill-spill geometry is already self-limiting. Heterogeneous by design; each cap cites its anchor in the parameter register.

2.5 Terrain

C10 — why not FABDEM. FABDEM is CC BY-NC — it would break the paper's central "license-clean, third-party-rebuildable" claim (and its Table I axis). The self-built correction ($z_{\text{DSM}} - b_0 - h_{\text{building}} - f \cdot h_{\text{canopy}}$, per-city fit to held-out ICESat-2 ATL08 + GEDI L2A, never transferred between cities) halves MAE and zeroes the urban-core bias where lidar is dense (Bangkok 1.43 → 0.78 m, bias +1.31 → 0.00). The two-parameter model cannot fix spatially-varying bias — accepted as screening-grade; the per-zone accuracy gate is the guard.

C11 — Singapore is the one city on a different terrain recipe (DeltaDTM low terrain + DSM-composited >30 m hills), because its ICESat-2/GEDI residual is 3–5 m MAE — scatter no per-city fit can beat. This is the "conditions under which more accurate terrain improves a risk surface" contribution *stated as a finding* rather than a buried inconsistency: the calibration is gated per city, and the city that fails the gate gets the best open substitute. DISCLOSED in the paper and §4 of the docs.

Subsidence correction only for BKK/JKT. Criterion: documented post-2013 InSAR rates (up to ~25 cm/yr north Jakarta, 1–2 cm/yr Bangkok); SG/KL have no comparable documented signal. Omission criterion now stated here (was implicit).

2.6 Defences (heterogeneity C9)

Seawall crests are burned per city from documented protection levels (SG +3.0 m, BKK +2.1 m BMA dike, JKT +2.0 m MSL; `scripts/_fix_coastal_seawall.py`). The pumped-polder *drain floor* is applied to **Bangkok only** (`apply_pumped_polder.py` in the release script), although Jakarta polder rings (Pluit, Muara Baru, Penjaringan) exist in the same module. *Rationale, previously code-comment-only, now on record:* Bangkok's banded CBD is documented to have stayed dry through 2011 within its design event (166 BMA pump stations, 80 mm/hr design) — the pump physics demonstrably hold; Jakarta's named polders have documented chronic rob flooding and pump under-capacity (Pluit inundated in 2020), so draining them would over-protect against the observed record. The asymmetry follows the evidence, not convenience: the defence is applied exactly where the defence is documented to work. Jakarta's gate (CRR 0.92) needs no polder drain to pass.

2.7 Validation statistics (C13, C15)

Bangkok's "significant skill" rests on the stratified-percentile bootstrap CI [0.19, 0.50] with a *degenerate* control stratum (7/7 correct rejections — zero resampling variance), while its exact Fisher test is marginal ($p \approx 0.08$). The abstract names the CI criterion inline and the table footnote carries the Fisher values, so the claim is precise as worded; both statistics must continue to travel together. Scoring every register at RP100 ("event-matched") is symmetric — controls are tested against the same footprint, so TSS punishes over-flooding — and register events are multi-year flood records rather than single events; the phrase could still be glossed at first use.

3. Cross-city heterogeneity register

Dimensi on	SG	KL	BKK	JKT	Why it differs	Risk
Terrain recipe	DeltaDTM + DSM-hills hybrid	lidar-calibrated bare-earth (ETH variant)	lidar-calibrated + subsidence	lidar-calibrated + subsidence	lidar density gates the calibration; documented InSAR rates gate subsidence	hybrid bias undisclosed at cell level (gated per zone)

Dimensi on	SG	KL	BKK	JKT	Why it differs	Risk
Coastal solver	inertial	none (domain ends ~7 km inland)	inertial	inertial	domain clipped to urban core	Port Klang fringe unscreened (disclosed)
MSL→EG M2008 offset	1.1588	n/a	1.1785	0.9976	per-gauge CMEMS MDT	dm-scale datum error on flat deltas
Defence: crest	+3.0 m	n/a	+2.1 m	+2.0 m	documented protection levels	single continuous crest per city (model-documentation §10 item 4)
Defence: pump floor	no	no	yes	no (rings exist, unused)	pumps documented to hold (BKK 2011) vs documented to fail (JKT rob)	asymmetry must cite evidence (now does)
Pluvial scheme	engine handfill	fill-spill u canal-HAND	fill-spill	engine handfill u canal-HAND	mechanism-matched (depression vs drainage-corridor)	union layers live in post-processors, not the engine flag
Handfill stage / cap	1.5 m / 0.5 m	0.5 m@0.095 ref / 3.0 m	— / 3.0 m	2.5 m engine + 0.3 m@0.130 canal / 1.0 m	per-city ponding anchors; scenario-scaled by IDF excess	base stages gate-visible (§10 item 3, OPEN)
Canal network pruning	n/a (engine HAND)	≥0.5 km ² accumulation	n/a	unpruned	KL hills have non-conveyance headwater ditches; JKT all-engineered (and gate-insensitive)	—
IDF anchor duration	1 h	6 h	6 h	6 h	national design-standard practice	KL 1-h variant untested
Runoff coefficient	0.75 (+raster)	0.75 (+raster)	raster only	0.80 (+raster)	WorldCover raster primary; scalar fallback per land-use mix	second-order
Fluvial HAND reference	main-stem v3	main-stem ≥180 km ² (v3eth)	debiased trunk	main-stem v3	same rule (trunk the discharge represents), per-city accumulation	—
Register size (pos/dry)	38/20	31/12	32/7	33/12	documented-record availability	BKK 7-control stratum degenerate (C13)
Manning n (coastal)	0.06	n/a	0.06	0.06	homogeneous — but never varied	C5 OPEN
Surge hydrograph	3-1-2 h	n/a	3-1-2 h	3-1-2 h	homogeneous — generic design event	C6 OPEN

The register shows the heterogeneity is of two kinds: **evidence-driven** (terrain recipe, pump floor, pruning, durations, caps — each anchored to a documented per-city fact) and **deliberately homogeneous simplifications** (Manning n, hydrograph shape) whose *lack* of per-city variation is itself the untested assumption. The first kind is defensible and now documented; the second kind is where the two OPEN sensitivity runs sit.

4. Actions taken this round

- 1 `ieee-kse2026.tex`: four corrections (C1 mitigation-delta mechanism → measured saturation numbers; C2 abstract bias mechanism; C3 stale pluvial limitation; C4 KL union parenthetical). SAFE v5's corresponding sentences were checked and are already direction-neutral ("mis-state") or correct — no SAFE edits needed.
- 2 This document written; summary folded into `model-documentation.md` §10.
- 3 Rationales that existed only as code comments (Jakarta polder non-application, bankfull upstream baking, cap anchors) promoted to the documentation record above.

5. Ranked follow-ups (pre-registered, none blocks the current submission)

- 1 **Coastal sensitivity bracket** (C5+C6, one afternoon): Bangkok 2100 RP100 with $n \in \{0.03, 0.10\}$ and $t_{\text{end}} = 16$ h; record extent deltas and per-cell converged/ t_{end} -limited flags in the parameter register.
- 2 **Held-out event validation** for the canal-HAND layer (C7): score one post-2022 documented flood absent from every register, before the next submission cycle.
- 3 **Physical re-anchoring of handfill base stages** (model-documentation §10 item 3, standing): drain design freeboard from PUB CoP / MSMA / PU standards.
- 4 **KL 1-h IDF variant** (C14): one pluvial re-fit to test duration sensitivity.
- 5 Gloss "event-matched RP100" at first use (C15) and, if a page frees up, restore the one-sentence forcing-uncertainty disclosure to KSE (present in SAFE v5).